

Nicholas Morello

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SUMMARY

Software engineer focused on building systems that turn ideas into working software. Hired as a backend engineer at Cox Automotive and moved onto the foundational AI Tech Software team, where I establish how AI-assisted workflows get built and adopted across the engineering org.

EXPERIENCE

Nextgear Capital

July 2025 – Present

Software Engineer - AI Tech

Indianapolis, IN

- **Self-initiated** the **agent harness** now used across Product & Engineering with no mandate behind it: **150+ engineers, 30+ contributors**, modular on-demand components over always-on integrations
- Runs **effectively as open source inside the org**: teams build, own, and share capabilities via self-serve pull requests, CI publishes to every engineer, privileged capabilities admin-gated
- Partnered with senior architects on **autonomous multi-agent workflows** carrying context and validation from spec to implementation to review
- Built the connective layer: **CLI tooling and on-demand skills** handing agents reliable context, **MCP integrations** into internal and external tools, plus **token and cost telemetry** on usage

Tech: Java, Python, Azure Functions, REST APIs, Microservices, Node.js, Claude Code, GitHub Copilot, Spec-Driven Development, Agentic Tooling

Raytheon Technologies

Feb. 2023 – July 2025

Embedded Software Engineer

Tucson, AZ

- Developed **embedded C++ software for mission-critical real-time systems**, ensuring real-time performance and reliability under strict aerospace standards
- Extensive **Python integration for firmware development**, working with hardware interfaces and automating validation workflows
- Optimized code for resource-constrained embedded platforms under aerospace certification requirements

Tech: C++, Python, RTOS, Embedded Systems

PROJECTS

Hue U | *AI-Powered Coloring Book Platform* | *Live at hueu.ai + App Store*

- **Directed AI coding agents** to architect, implement, test, and ship this full-stack platform (web + native iOS) end-to-end, backed by full CI/CD and test coverage
- Engineered the **AI photo-to-line-art** pipeline and an interactive book builder with print-on-demand fulfillment and a **Stripe-powered credit system**
- Built a **native iOS app** (SwiftUI) **shipped to the App Store**, with a custom flood-fill coloring engine, Apple Pencil pressure support, and StoreKit 2 in-app purchases
- Built an **autonomous Hermes agent** for production operations, with approval-gated write actions behind a human-in-the-loop gate

Tech: Python, FastAPI, Next.js, Swift (iOS), Supabase, Google Gemini, Stripe, Docker, Google Cloud, Hermes, Agentic Development

EDUCATION

Arizona State University

Bachelor of Science in Computer Science

SKILLS

Languages : Java, C++, Python, Swift, JS / TS

Frameworks & Tools : React, Next.js, Python FastAPI, Azure, Supabase, MongoDB, Docker, REST APIs

AI & Methodology : Claude Code, Codex, Custom Pi Agent, Hermes Agent, Agent Skills & CLI Tooling, MCP, Multi-Agent Workflows, Spec-Driven Development